

Sowhardo Bin Atik

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[GitHub](#) · [Portfolio](#) · [LinkedIn](#)

Objective

Full-Stack Developer with deep experience in Node.js/Express backends and PostgreSQL database design. Proven track record in implementing secure JWT authentication, Docker containerization, and RAG-based AI integrations. Looking to contribute strong system optimization, Development and AI integration skills.

Education

BRAC University 2022 - Expected June 2026
Bachelor of Science in Computer Science and Engineering CGPA: 3.69 / 4.00
Coursework completed; thesis defense scheduled for 13th June 2026

Technical Skills

Languages: JavaScript, Python, Java, C, C++, C#, TypeScript (Learning)
Backend & Networks: Node.js, Express.js, RESTful APIs, RAG, Redis, Nginx, HTTPS, SSL/TLS, DNS, TCP/IP
Databases & ORMs: pgvector, PostgreSQL, MongoDB, ACID, Prisma (Learning)
DevOps & Testing: Docker, Linux, Bash, Git, Github Actions, SSH, Vim, Virtual Machines, Jest, Autocannon
Frontend & AI Tools: React.js, Tailwind CSS, Figma, Claude (Skills & Artifacts), GitHub Copilot, Cursor
Problem Solving: LeetCode [\[Profile\]](#)

Projects

MetroSheba: AI-Driven Transit & Ticketing Platform [\[Live\]](#) · [\[Video\]](#) · [\[GitHub\]](#)
Technologies: Express.js, Node.js, React, PostgreSQL, Redis, RAG

- Developed a full-stack web application featuring a **real-time station dashboard** that tracks passenger crowds; modifications made by system managers are instantly broadcast live to client displays.
- Integrated a smart AI chat assistant using **vector embeddings for semantic searches**, implementing an optimized **Redis cloud caching layer** to serve repeat queries instantly.
- Engineered an automated ticketing engine secured by a **dual-token authentication system** (Access/Refresh tokens) via **HttpOnly cookies** for safe session management.
- Set up automated testing suites using **Jest**, integrating an automated **GitHub Actions CI/CD workflow** to validate code reliability on every push.
- Conducted **stress and load testing** to evaluate application resilience, successfully identifying bottlenecks and tuning **connection-pooled queries** under heavy traffic.
- Utilized **AI-assisted workflows** to minimize boilerplate code and proactively catch syntax bugs during development.

Multi-Agent Research Assistant [\[GitHub\]](#)
Technologies: FastAPI, Python, React, Groq API, Asynchronous Programming

- Developed an **asynchronous modular backend** using FastAPI to handle concurrent task delegation and real-time response aggregation across **multiple LLM agents**.
- Implemented structured data synthesis pipelines to automate cross-domain research fetching and parsing into **standardized client payloads**.
- Developed a lightweight React frontend to interface with the agent execution engine via **optimized REST endpoints**.

Remote Deployment Pipeline [\[Video\]](#) · [\[GitHub\]](#)
Technologies: Express.js, Bash, OpenSSH, Linux

- Implemented a **single-command deployment pipeline** using Bash and SCP to securely sync source code with a remote Linux VM via **Ed25519 key-based authentication**.

Achievements

AI CERTs Prompt Engineer Level 1 Certification 2024
Learned LLM architecture fundamentals to apply advanced prompting methodologies (zero/few-shot, Chain-of-Thought, and Role prompting), significantly improving software development lifecycles and AI integration efficiency. [\[View Certificate\]](#)